

Nitin Sharma

Delhi, India — +91-9205646560 — nitin18geo@gmail.com — [linkedin.com/in/nitingeology](https://www.linkedin.com/in/nitingeology)

Education

Indian Institute of Technology Kharagpur <i>M.Sc. Geology, GPA: 8.95/10</i>	Kharagpur, India 2021–2023
National Law University & World Wide Forum India <i>P.G. Diploma in Environmental Law and Policy (Grade A)</i>	Remote 2022–2023
Hansraj College, University of Delhi <i>B.Sc. (Hons.) Geology, GPA: 8.65/10</i>	Delhi, India 2018–2021

Experience

School of Public Policy, IIT Delhi <i>Young Professional</i>	Delhi, India Feb 2025–Present
– Synthesized Earth observation, survey and hydrological datasets to evaluate equity, efficiency and sustainability of irrigation water use in collaboration with IHE Delft. – Contributed to a decision support system and coordinated interdisciplinary teams developing a system dynamics model for Varuna River rejuvenation under the National Mission for Clean Ganga.	
International Water Management Institute (IWMI) <i>Research Consultant</i>	Delhi, India Jan 2024–Dec 2024
– Evaluated Direct Seeded Rice adoption in Haryana using satellite imagery, field surveys and machine learning in Google Earth Engine. – Developed reproducible Earth observation pipelines for drought monitoring in Africa and automated flood inundation mapping supporting the International Disaster Charter.	
International Water Management Institute (IWMI) <i>Research Intern</i>	Delhi, India Jun 2023–Dec 2023
– Developed a Google Earth Engine workflow for monitoring surface water dynamics using Landsat and Sentinel imagery; delivered tools to the Central Water Commission and conducted GEE training workshops.	
Arizona State University <i>Research Intern</i>	Remote Jun 2023–Aug 2023
– Developed a machine learning workflow for mapping Animal Feeding Operations to support watershed nutrient runoff and environmental policy analyses.	

Skills

Programming: Python, R
Geospatial Analysis: Google Earth Engine, Remote Sensing, GIS, Spatial Analysis
Machine Learning: Classification, Validation, Model Development
Visualization: QGIS, ArcGIS StoryMaps, Dashboards
Policy: Decision Support Systems, Evidence Synthesis, Science Communication

Publications & Reports

Journal: Patterns of Surface Water Dynamics and Storage Changes in Bundelkhand Region, India. *Journal of Irrigation and Drainage*, 2024. DOI
Report: Building Resilient River Basins with the Catchment Risk and Resilience (CRI) Index. CGIAR Repository
Article: Effect of Rainfall Deficit on Surface Water Storage Changes. CGIAR Article — StoryMap
Conference: Mapping Surface Water in Un-monitored Small Waterbodies Using Earth Observation Data, AGU Hydrology 2024. Abstract

Awards

Best StoryMaps Award: Esri India StoryMaps Contest, 2022. [Link](#)
L. Austin Weeks Undergraduate Grant: 2020